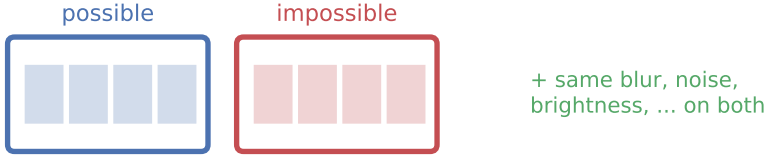
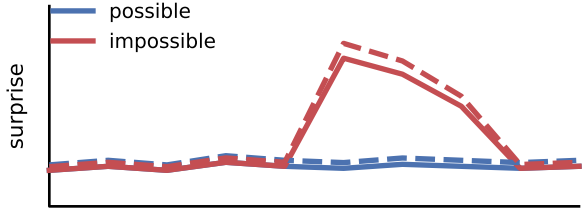
Violation-of-expectation: a model is 'surprised' by impossible videos. What is that surprise tracking?

1 Invariance

Does a harmless change to the video change which one looks impossible?

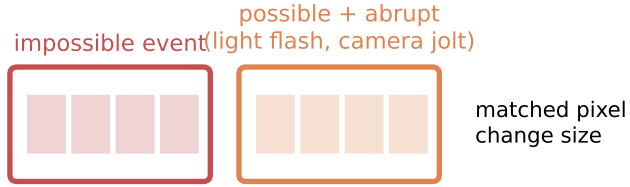


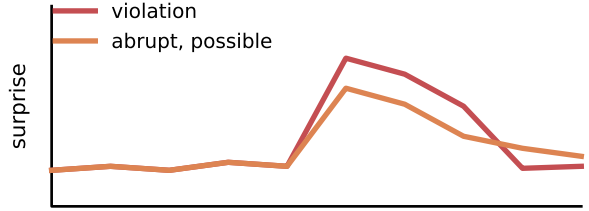


same verdict before (solid) and after (dashed)?

2 Specificity

Is surprise about impossible physics, or about any sudden change?

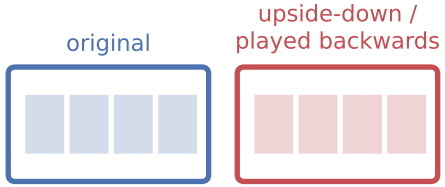


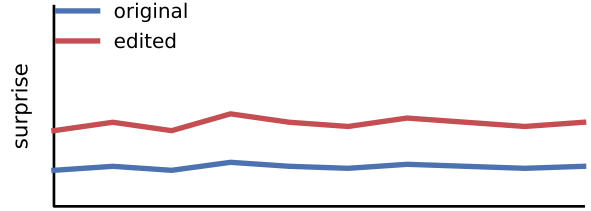


does the model tell them apart?

3 Positive controls

If we make the physics impossible, does surprise go up?





checks that the pipeline can see violations